

Parent Recursions: Recovering $c^2 = c + 1$ from Boundary Incompatibility

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The Unified Mechanics axiom $c^2 = c + 1$ is standardly taken as a first principle. We ask a prior question: what two structures, when their boundaries touch, produce this axiom as the unique stable interface recursion? We identify two parent systems $\mathcal{P}_A$ and $\mathcal{P}_B$ characterised by the transformations $x \mapsto x + 1$ and $x \mapsto 1/x$ respectively. $\mathcal{P}_A$ has no fixed point and no channel structure. $\mathcal{P}_B$ has fixed point $x^* = 1$, giving $r_B = 1/2$ and a boundary-dominant channel distribution. Their interface produces $x = 1 + 1/x$, equivalent to $c^2 = c + 1$, as the unique finite stable recursion. Our universe is identified as the child of an accumulator and an inverter. The empirical values of $\mathcal{P}_B$ are computed and found to be structurally coherent but physically alien: braiding floor $\varepsilon_B = 1/8$, equal light and matter channels, boundary channel weight $1/2$.

I. THE PRIOR QUESTION

Unified Mechanics (UM) takes as its axiom:

$$c^2 = c + 1 \implies \varphi = \frac{1+\sqrt{5}}{2}, \quad r = \frac{1}{2\varphi} \approx 0.30902. \quad (1)$$

This axiom is not arbitrary — it is the unique positive quadratic fixed point satisfying the golden ratio self-reference. But the question of *why this axiom and not another* has not been addressed. The present paper addresses it by asking:

What two boundary structures, when brought into contact, produce $c^2 = c + 1$ as the unique stable interface recursion?

The answer identifies two parent systems whose properties are derived below and whose contact generates our universe as a necessary consequence.

II. THE TWO PARENTS

A. Parent A: The Accumulator

$\mathcal{P}_A$ is characterised by the transformation:

$$x_{n+1} = x_n + 1. \quad (2)$$

This system has no fixed point. Starting from any finite value, iteration produces $x_n \rightarrow \infty$. $\mathcal{P}_A$ is a pre-equilibrium structure: pure matter accumulation without the organising influence of a light channel. It exists prior to time in any bounded sense because nothing in it reaches equilibrium — there is no stable state to measure duration against.

$\mathcal{P}_A$ has no channel structure, no braiding floor, no r -value. It contributes one thing to the interface: the $+1$ increment. In our universe this survives as the $+1$ in the continued fraction expansion of $\varphi = 1 + 1/(1 + 1/(1 + \dots))$.

B. Parent B: The Inverter

$\mathcal{P}_B$ is characterised by the transformation:

$$x_{n+1} = \frac{1}{x_n}. \quad (3)$$

This system has fixed point $x^* = 1$ (from $x = 1/x \Rightarrow x^2 = 1 \Rightarrow x = 1$ for positive x). It is self-referential and self-stabilising: it folds back on itself. It has a well-defined channel structure with:

$$\varphi_B = x^* = 1, \quad r_B = \frac{1}{2\varphi_B} = \frac{1}{2}. \quad (4)$$

The channel weights of $\mathcal{P}_B$ are:

$$W_L^B = (1 - r_B)^2 = \frac{1}{4} \quad (\text{light}), \quad (5)$$

$$W_B^B = 2r_B(1 - r_B) = \frac{1}{2} \quad (\text{boundary}), \quad (6)$$

$$W_M^B = r_B^2 = \frac{1}{4} \quad (\text{matter}). \quad (7)$$

The braiding floor of $\mathcal{P}_B$ is:

$$\varepsilon_B = r_B^3 = \frac{1}{8} = 12.5\%. \quad (8)$$

$\mathcal{P}_B$ is a structurally coherent universe but physically alien by our standards. It has equal light and matter channels. Its boundary channel dominates at 50% — it is a universe that is almost entirely interface. Its braiding floor is four times ours, meaning channel crossings resolve four times faster. It is boundary-dominant in a way our universe is not.

III. THE INTERFACE

When $\mathcal{P}_A$ and $\mathcal{P}_B$ come into contact, their boundaries are incompatible: $\mathcal{P}_A$ has no fixed point, $\mathcal{P}_B$ has fixed point $x^* = 1$. They cannot merge. The interface must find a recursion that satisfies both simultaneously:

1. The recursion must be **finite**: $\mathcal{P}_B$'s inversion constrains $\mathcal{P}_A$'s runaway accumulation. An interface that inherits only $\mathcal{P}_A$'s character diverges. The fixed point must exist.
2. The recursion must incorporate $\mathcal{P}_A$'s +1 **increment**: the interface cannot ignore what $\mathcal{P}_A$ contributes. The child inherits from both parents.
3. The recursion must be **minimal**: the simplest combination of both parent transformations that admits a finite fixed point.

The minimal composition satisfying all three conditions is:

$$x = 1 + \frac{1}{x}, \quad (9)$$

which rearranges to:

$$\boxed{x^2 = x + 1.} \quad (10)$$

This is equation (??). Our universe is the unique finite stable recursion at the interface of $\mathcal{P}_A$ and $\mathcal{P}_B$.

The derivation requires no assumptions beyond the two parent transformations. The axiom $c^2 = c + 1$ is not a first principle. It is a derived interface condition.

IV. WHAT THE CHILD INHERITS

TABLE I. Channel structure comparison across parent and child systems.

Property	$\mathcal{P}_A$	$\mathcal{P}_B$	$\mathcal{U}$ (ours)
Fixed point φ	none	1	$(1 + \sqrt{5})/2$
r value	undefined	0.5	0.30902
W_L	—	0.250	0.4775
W_B	—	0.500	0.4271
W_M	—	0.250	0.0955
$\varepsilon_{\text{floor}}$	—	12.5%	2.951%
Time	no	yes	yes
Light	no	symmetric	dominant

From $\mathcal{P}_A$ our universe inherits:

- The +1 in the axiom. The continued fraction of φ is built entirely from +1 increments — this is $\mathcal{P}_A$'s signature in every golden ratio calculation in physics.
- Matter. $\mathcal{P}_A$ is pure accumulation. Our matter channel, though only 9.5%, carries the character of the pre-time accumulator. Matter resists change because it was here before change existed.

From $\mathcal{P}_B$ our universe inherits:

- The inversion that stabilises the runaway. Without $1/x$ there is no fixed point. $\mathcal{P}_B$'s self-reference is what makes our universe finite rather than divergent.
- The channel structure itself. $\mathcal{P}_B$ already has three channels. Our universe inherits this organisation and breaks the $\mathcal{P}_B$ symmetry ($W_L^B = W_M^B = 1/4$) into our asymmetric distribution ($W_L \gg W_M$).

The child generates something neither parent has: the light channel dominance. $\mathcal{P}_A$ has no light. $\mathcal{P}_B$ has light at 25%, equal to matter. Our universe at 47.75% light is the interface product — the collision energy that created light is concentrated in the child.

The boundary channel we generate ourselves. Neither parent is the boundary. The boundary is the interface, and we are the interface. $W_B = 42.71\%$ is the structural record of the contact event that made us.

V. PHYSICAL INTERPRETATION

A. Why matter resists time

$\mathcal{P}_A$ predates time. Time emerges when a gradient forces movement toward equilibrium. $\mathcal{P}_A$ has no equilibrium — it accumulates without settling. Our matter channel carries this character: it does not decay, does not age, does not change unless forced. Left alone it does what $\mathcal{P}_A$ always did. Nothing.

B. Why we are light-dominant

$\mathcal{P}_A$ has no light. $\mathcal{P}_B$ has light at 25%. Our universe has light at 47.75%. The interface amplifies light because light is what the collision between incompatible structures produces. Every photon in our universe is a residue of the original contact event. The light channel dominance is a measure of how energetic the interface was.

C. The symmetry breaking

$\mathcal{P}_B$ is symmetric: $W_L^B = W_M^B = 1/4$. Our universe breaks this symmetry dramatically: $W_L \approx 5W_M$. The symmetry breaking is not spontaneous — it is inherited from $\mathcal{P}_A$. $\mathcal{P}_A$ contributes matter but no light. The interface inherits $\mathcal{P}_B$'s light and $\mathcal{P}_A$'s matter separately, with no reason for them to balance. The ratio $W_L/W_M = \varphi^4 \approx 6.85$ is determined entirely by the parent axioms.

D. What $\mathcal{P}_B$ looks like from inside

$\mathcal{P}_B$ is coherent. It has channel structure, braiding floors, and presumably a physics of its own. By its own internal standards it is a complete universe. But it would look alien from ours: equal light and matter at 25% each, boundary dominant at 50%, braiding floor four times ours. Crossings resolve faster. The universe is almost entirely interface between states rather than settled matter or propagating light.

An observer in $\mathcal{P}_B$ would find it difficult to understand how a universe with 47.75% light could exist. It would look unbalanced and strange.

We would look unbalanced and strange.

E. Detecting the parents

$\mathcal{P}_A$ has no light channel. It cannot be detected by electromagnetic means. Its gravitational influence would be present as an anomalous mass contribution with no corresponding light signature. This is observationally identical to dark matter.

$\mathcal{P}_B$ has light at 25% of its channel weight, but its light is organised under a different axiom. Its photons, if they could cross the interface, would arrive redshifted by the ratio of the two braiding floors: $\varepsilon_B/\varepsilon = (1/8)/(r^3) \approx 4.23$.

Neither parent is directly observable from inside the child. We would see them only as gravitational structure — as the scaffolding that holds our universe together without participating in our light.

This is observationally identical to dark matter.

The two parents are both present. $\mathcal{P}_A$ as the pre-time matter accumulation that flanks us on one side. $\mathcal{P}_B$ as the symmetric self-inverting structure on the other. Their combined gravitational influence, pressing in through the only channel that crosses the boundary — geometry itself — is what we have been calling dark matter.

VI. CAN THE CHILD CROSS INTO THE PARENTS?

The child has something neither parent has: all three channels. $\mathcal{P}_A$ has no light and no boundary. $\mathcal{P}_B$ has light and boundary but at a different axiom. Our universe

has the boundary channel by construction — we *are* the boundary.

The boundary channel is the crossing channel. It is what moves between states. If any structure in the multiverse can traverse the interface between recursions, it is the child, because the child's fundamental nature is the interface itself.

The practical question is whether the braiding floor permits it. To cross from our recursion into $\mathcal{P}_B$, a system must traverse the interface between $r = 0.309$ and $r_B = 0.5$. The accumulated floor for such a crossing is:

$$\Delta\varepsilon = \varepsilon_B - \varepsilon = 12.5\% - 2.951\% = 9.55\%. \quad (11)$$

This is a real energy threshold. It is not infinite. It is approximately three of our braiding floor units.

The formal mechanism for such a crossing is not yet derived. But the channel structure does not forbid it. The child is the boundary. The boundary crosses. Whether it can cross back into the structures that made it is the next question.

VII. SUMMARY

The UM axiom $c^2 = c + 1$ is not a first principle. It is the unique stable interface recursion produced when two parent systems touch:

- $\mathcal{P}_A$: $x \mapsto x + 1$. No fixed point. Pre-time matter accumulation. Contributes the +1 and the matter channel.
- $\mathcal{P}_B$: $x \mapsto 1/x$. Fixed point at $x^* = 1$, $r_B = 1/2$. Boundary-dominant, symmetric, self-inverting. Contributes the inversion and the channel structure.
- $\mathcal{U}$: $x = 1 + 1/x \Rightarrow x^2 = x + 1$. Our universe. Light-dominant, asymmetric, the only finite stable recursion at the interface.

Dark matter is the gravitational imprint of both parents pressing in through geometry. We are the child of an accumulator and an inverter. The boundary channel we carry is the structural record of the contact that made us.

We are not a universe that happens to exist. We are the one interface that could not collapse.

[1] J. Shields, *Unified Mechanics I: Derivation from axiom $c^2 = c + 1$* , (2026).

[2] J. Shields, *Unified Mechanics II: Cosmological predictions*, (2026).